SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

QUESTIONS: 5 **Max Marks50** **Duration :60 Minutes Annexure A**

Scenario-Based Questions

Code of Conduct:

I Anto Princy A/ 192411071 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Scenario-Based Questions

1. Document Image Enhancement and Text Extraction

A library digitises old manuscripts and printed records. The scanned images suffer from faded ink, yellowed and unevenly lit backgrounds, salt-and-pepper noise, and ink bleed-through from the reverse side. The system must separate the text from the background before character recognition.

Question: Apply suitable grey-level transformations, histogram processing, and spatial filtering to improve the legibility of the scanned pages. Subsequently, apply an appropriate global, adaptive, or optimal thresholding technique to extract the text regions. Explain the sequence of operations and justify why a single global threshold may or may not be adequate for this image.

1. Document Image Enhancement and Text Extraction

Problem Analysis

The scanned manuscript suffers from multiple degradations:

- Faded ink causing low text contrast.
- Uneven illumination creating varying background brightness.
- Salt-and-pepper noise introducing random black and white pixels.
- Ink bleed-through from the reverse side producing unwanted text-like patterns.

A single processing technique cannot handle all these issues effectively. Therefore, a sequence of enhancement and segmentation operations is required before OCR (Optical Character Recognition).

Step 1: Grey-Level Transformations

Contrast Stretching

Enhance the difference between faded text and background.

$$s = \frac{(r - r_min)}{(r_max - r_min)} \times 255$$

Purpose:

- Makes faded characters darker.
- Expands the intensity range of the image.
- Improves visibility of weak text strokes.

Gamma Correction (if image is dark)

$$s = cr^{\gamma}$$

- Choose $\gamma < 1$to brighten dark regions.
- Helps reveal hidden text details.

Result: Improved text contrast and readability.

Step 2: Histogram Processing

Adaptive Histogram Equalization (CLAHE)

Instead of global histogram equalization, apply Contrast Limited Adaptive Histogram Equalization (CLAHE).

Why?

- Old manuscripts often have non-uniform lighting.
- Different image regions require different contrast enhancement.

Advantages:

- Enhances local text details.
- Prevents over-amplification of noise.
- Improves separation between text and paper background.

Result: Better visibility of text in both bright and dark regions.

Step 3: Spatial Filtering

(a) Median Filtering

Use a **3×3 or 5×5 median filter**.

Purpose:

- Removes salt-and-pepper noise.
- Preserves character edges.

Why not mean filter?

- Mean filtering blurs text boundaries.
- Median filtering effectively removes impulse noise while maintaining sharpness.

(b) Morphological Operations

Opening (Erosion + Dilation)

Removes small isolated noise pixels.

Top-Hat Transformation

$$\text{Top-Hat} = f - (f \circ b)$$

where f is the image and b is the structuring element.

Purpose:

- Highlights dark text on bright background.
- Suppresses uneven illumination.

(c) Background Suppression

To reduce bleed-through:

- Apply large-kernel Gaussian smoothing to estimate the background.
- Subtract the estimated background from the original image.

Result:

- Reverse-side ink becomes less prominent.
- Foreground text becomes dominant.

Step 4: Thresholding for Text Extraction

Recommended Method: Adaptive Thresholding

Examples:

- Adaptive Mean Thresholding
- Adaptive Gaussian Thresholding
- Sauvola's Method
- Niblack's Method

Reason: Background intensity varies across the manuscript.

For each local window:

$$T(x, y) = \text{Local Mean} - C$$

Pixels are classified as:

$$g(x, y) = \begin{cases} 1, \& f(x, y) > T(x, y) \\ 0, \& f(x, y) \leq T(x, y) \end{cases}$$

- Text is extracted accurately from both bright and dark areas.
- Handles illumination variations effectively.

Why a Single Global Threshold May Not Be Adequate?

A global threshold uses one value T for the entire image:

$$g(x, y) = \begin{cases} 1, \& f(x, y) > T \\ 0, \& f(x, y) \leq T \end{cases}$$

Problems:

1. **Uneven Illumination**

- Some background regions are darker than others.
- One threshold cannot correctly separate all text pixels.

2. **Faded Characters**
    - Weak text may have intensities close to the background.
    - Global threshold may remove them.
3. **Bleed-Through Marks**
    - Reverse-side ink may be incorrectly classified as foreground text.
4. **Local Contrast Variation**
    - Different page areas have different intensity distributions.
    - A single threshold ignores these local differences.

When Global Thresholding Works

- Uniform lighting.
- High text-to-background contrast.
- Minimal noise and degradation.

Since the manuscript contains severe illumination variation, fading, and bleed-through, **adaptive thresholding is superior to global thresholding**.

Recommended Processing Sequence

Scanned Document

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Contrast Stretching / Gamma Correction

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CLAHE (Adaptive Histogram Equalization)

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Median Filtering

↓

Morphological Opening + Top-Hat Transform

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Background Estimation and Subtraction

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Adaptive Thresholding (Sauvola/Niblack)

↓

Binary Text Image

↓

OCR (Character Recognition)

To enhance the degraded manuscript, apply **contrast stretching and CLAHE** to improve text visibility, followed by **median filtering** and **morphological operations** to remove salt-and-pepper noise, uneven illumination, and bleed-through effects. Finally, use **adaptive thresholding (preferably Sauvola or Adaptive Gaussian)** for text extraction. A **single global threshold is not suitable** because the image contains non-uniform illumination, faded text, and varying local contrast, which require locally adaptive decision-making for accurate segmentation.

2. Printed Circuit Board (PCB) Fault Detection

An electronics manufacturer inspects PCBs using a high-resolution camera. The images contain fine copper tracks, solder joints, and component pads. Faults such as broken tracks, hairline cracks, short circuits, and missing solder points appear with low contrast and are affected by reflections and sensor noise.

Question: Apply appropriate smoothing and sharpening techniques to enhance the PCB images. Use point detection, line detection, and edge detection operators to locate potential faults. Apply edge linking and boundary detection to obtain continuous track outlines and defect contours. Describe the processing pipeline and the contribution of each technique to fault identification.

Printed Circuit Board (PCB) Fault Detection

Introduction

In modern electronics manufacturing, Printed Circuit Boards (PCBs) are inspected using high-resolution imaging systems to ensure product quality and reliability. A PCB contains intricate copper tracks, solder joints, vias, and component pads that must be free from defects to guarantee proper electrical connectivity. However, manufacturing defects such as broken tracks, hairline cracks, short circuits, missing solder points, and track width variations may occur during fabrication and assembly. These defects are often difficult to detect because they appear with very low contrast and may be obscured by reflections, uneven illumination, or sensor noise.

To accurately identify these defects, a sequence of digital image processing techniques is applied. The process includes image smoothing, sharpening, point detection, line detection, edge detection, edge linking, and boundary detection. Each technique contributes to highlighting different PCB features and improving defect visibility. The final result is a precise identification of faulty regions for automated quality inspection and classification.

Step 1: Image Smoothing (Noise Reduction)

Before detecting defects, unwanted noise and image artifacts must be removed. Noise can create false edges and mislead subsequent processing stages.

Gaussian Filtering

A Gaussian filter smooths the image by averaging neighbouring pixel values according to a Gaussian distribution:

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

The Gaussian filter effectively reduces random sensor noise while preserving the overall structure of copper traces and solder joints.

Median Filtering

A median filter replaces the centre pixel with the median value of its surrounding neighbourhood:

$$f(x, y) = Median(N)$$

where Nrepresents the local neighbourhood.

Advantages of Median Filtering:

- Removes salt-and-pepper noise effectively.
- Preserves important edges of PCB tracks.
- Maintains structural details better than averaging filters.
- Prevents blurring of thin conductive paths.

Contribution to Fault Detection

The smoothing stage helps eliminate noise-induced irregularities and reflection artifacts. As a result, the PCB image becomes cleaner, allowing later stages to focus only on genuine defects rather than noise-generated features.

Step 2: Image Sharpening

After smoothing, some fine details may become less distinct. Sharpening is therefore applied to restore and enhance PCB features such as thin copper traces, solder boundaries, and small cracks.

Laplacian Sharpening

The Laplacian operator is a second-order derivative that highlights rapid intensity changes:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} \backslash + \frac{\partial^2 f}{\partial y^2}$$

The sharpened image is obtained using:

$$g(x, y) = f(x, y) - \nabla^2 f(x, y)$$

Effects

- Enhances fine structural details.
- Highlights defect boundaries.

- Improves track visibility.

Unsharp Masking and High-Boost Filtering

A blurred version of the image is first obtained and then subtracted from the original:

$$Mask = f - \bar{f}$$

The enhanced image becomes:

$$g = f + k(f - \bar{f})$$

where $k$is a gain factor.

Effects

- Makes narrow tracks more visible.
- Enhances weak cracks and breakpoints.
- Improves contrast of solder joints and pads.

Contribution to Fault Detection

Sharpening increases the distinction between normal PCB structures and defective regions, making subsequent detection operations significantly more reliable.

Step 3: Point Detection

Point detection is used to identify isolated defects that occupy only a few pixels and differ significantly from their surroundings.

Point Detection Mask

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

When the response exceeds a predefined threshold, the corresponding location is marked as a potential defect.

Defects Detected

- Missing solder points

- Pinholes in copper tracks
- Burn marks
- Tiny surface contaminants
- Isolated manufacturing defects

Contribution to Fault Detection

Point detection allows early localisation of very small defects that may not be identifiable using broader region-based methods.

Step 4: Line Detection

Copper traces are essentially line structures. Therefore, line detection operators are used to extract them and analyse their continuity.

Horizontal Line Mask

$$\begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

Vertical Line Mask

$$\begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$$

Diagonal Line Mask

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

Contribution to Fault Detection

Line detection helps:

- Extract copper trace networks.
- Detect missing segments.
- Identify broken conductive paths.

- Locate disconnected regions.
- Analyse track continuity throughout the PCB.

Broken tracks usually appear as interruptions in detected line patterns and can therefore be identified automatically.

Step 5: Edge Detection

Edges represent changes in intensity and correspond to boundaries of tracks, solder joints, cracks, and pads.

Canny Edge Detector

The Canny operator is widely preferred because it provides excellent detection accuracy and noise immunity.

Processing Stages

1. Gaussian smoothing
2. Gradient computation
3. Non-maximum suppression
4. Double thresholding
5. Edge tracking by hysteresis

The gradient magnitude is calculated as:

$$M = \sqrt{G_x^2 + G_y^2}$$

where:

- G_x= horizontal gradient
- G_y= vertical gradient

Advantages

- Detects weak defect edges.
- Produces thin and accurate boundaries.

- Minimises false detections.
- Preserves important PCB structures.

Contribution to Fault Detection

The edge detection stage identifies:

- Hairline cracks
- Broken tracks
- Solder pad boundaries
- Short-circuit bridges
- Defect contours

The resulting edge map forms the basis for further analysis and defect localisation.

- **Step 6: Edge Linking**

After edge detection, edges may appear fragmented because of noise, reflections, or low contrast. These disconnected pieces must be connected to form complete defect boundaries.

Hysteresis Edge Linking

Strong edges are retained automatically, while weak edges connected to strong edges are preserved.

Benefits

- Eliminates isolated false edges.
- Maintains genuine PCB boundaries.
- Connects weak crack segments.

Morphological Dilation

The dilation operation expands edge pixels:

$$A \oplus B$$

where $A$is the image and $B$is the structuring element.

Benefits

- Bridges small gaps between edge segments.
- Connects broken contours.
- Produces continuous track outlines.

Contribution to Fault Detection

Edge linking improves structural continuity, enabling more accurate analysis of conductor paths, cracks, and solder-joint boundaries.

- **Step 7: Boundary Detection and Contour Extraction**

Once continuous edges are obtained, boundary detection extracts the outlines of PCB components and defects.

Contour Tracing

Contour tracing follows connected boundary pixels and generates complete outlines of PCB objects.

Information Obtained

- Area
- Perimeter
- Length
- Width
- Shape descriptors
- Geometric properties

Chain Code Representation

Boundary directions are represented using numerical codes corresponding to neighbouring pixels.

Benefits

- Efficient boundary storage.
- Simple shape analysis.
- Accurate contour representation.

Region Boundary Analysis

The extracted contours can be compared with standard PCB templates.

Enables Detection Of

- Missing solder pads
- Excess solder accumulation
- Deformed conductive tracks
- Broken traces
- Unexpected copper bridges
- Incorrect component footprints

Contribution to Fault Detection

Boundary analysis provides quantitative measurements that help distinguish between acceptable manufacturing variations and actual defects.

Fault Identification Using Various Techniques

Defect Type	Detection Method
Broken Tracks	Line Detection + Edge Detection + Boundary Analysis
Hairline Cracks	Sharpening + Canny Edge Detection
Short Circuits	Edge Linking + Boundary Detection
Missing Pads	Point Detection + Contour Analysis
Missing Solder	Point Detection + Shape Analysis
Track Width Variations	Boundary Measurement
Copper Discontinuities	Line Detection + Contour Tracking
Excess Solder Bridges	Edge Detection + Boundary Analysis

3. Underwater Image Restoration and Object Segmentation

A marine research institute captures images of coral reefs and underwater structures. The images show a bluish-green colour cast, low contrast, haze due to scattering, and non-uniform illumination. Researchers need to extract coral and object regions for analysis.

Question: Apply suitable histogram processing and frequency-domain enhancement (such as homomorphic filtering) to correct the illumination and improve contrast. Select an appropriate low-pass or high-pass filter based on the degradation characteristics, and justify the cutoff selection. Then apply thresholding or region-based segmentation to isolate the target regions. Explain the order of operations and why that order is important.

Underwater Image Restoration and Object Segmentation

Introduction

Underwater imaging plays a vital role in marine research, coral reef monitoring, underwater archaeology, and environmental studies. However, images captured underwater often suffer from several degradations caused by the aquatic environment. Water absorbs and scatters light, resulting in a bluish-green colour cast, reduced contrast, haze effects, blurred object boundaries, and non-uniform illumination. These degradations make it difficult for researchers to accurately identify and analyse corals, marine organisms, rocks, and underwater structures.

To obtain meaningful information from underwater images, image restoration and enhancement techniques must first be applied. Histogram processing and frequency-domain enhancement methods such as homomorphic filtering can significantly improve image quality by correcting illumination variations and enhancing contrast. Once the image quality is improved, segmentation techniques such as thresholding or region-based segmentation can be used to isolate coral reefs and other underwater objects from the background. The order of these operations is crucial because segmentation accuracy depends heavily on the quality of the enhanced image.

Step 1: Colour and Contrast Enhancement Using Histogram Processing

Since underwater images usually exhibit poor contrast and colour distortion, histogram-based enhancement is first performed to improve visibility.

Histogram Equalization

Histogram equalization redistributes the intensity values of the image to utilise the full dynamic range.

Principle

The transformation function is:

$$s = T(r)$$

where:

- r= input intensity
- s= output intensity

The method spreads frequently occurring intensity levels and improves global image contrast.

Advantages

- Enhances dark regions.
- Improves visibility of underwater structures.
- Makes object boundaries more distinguishable.
- Increases the dynamic range of pixel intensities.

Limitation

Global histogram equalization may over-enhance certain regions and cannot effectively handle non-uniform illumination.

Contrast Limited Adaptive Histogram Equalization (CLAHE)

CLAHE performs histogram equalization locally on small image blocks.

Advantages

- Enhances local contrast.
- Preserves details in coral textures.
- Prevents excessive amplification of noise.
- Handles varying illumination across the image.

Contribution

CLAHE is particularly suitable for underwater images because different regions may receive different amounts of light due to water depth and scattering effects.

Step 2: Frequency-Domain Enhancement Using Homomorphic Filtering

After initial contrast enhancement, illumination effects and haze must be reduced. This is achieved using homomorphic filtering in the frequency domain.

Image Illumination Model

An image can be represented as:

$$f(x,y) = i(x,y) \cdot r(x,y)$$

where:

- $i(x,y)$= illumination component
- $r(x,y)$= reflectance component

Characteristics

- Illumination corresponds to low-frequency components.
- Reflectance corresponds to high-frequency components.

Underwater images generally have uneven illumination and haze, which dominate the low-frequency region.

Homomorphic Filtering Procedure

Step 1: Take Logarithm

$$\ln f\,(x,y) = \ln i\,(x,y) + \ln r\,(x,y)$$

This converts multiplication into addition.

Step 2: Apply Fourier Transform

$$F(u,v) = \mathcal{F}f(x,y)$$

The image is transformed into the frequency domain.

Step 3: Apply Homomorphic Filter

A high-pass filter is applied:

H(u,v)= (\gamma_H-\gamma_L) \left\[1-e^{-c(D^2(u,v)/D_0^2)}\right] +\gamma_L

where:

- D_0= cutoff frequency
- $\gamma_H > 1$
- $\gamma_L < 1$

Step 4: Apply Inverse Transform

The enhanced image is obtained by inverse Fourier transformation.

Why High-Pass Filtering is Preferred?

The degradation characteristics indicate:

- Non-uniform illumination
- Low contrast
- Haze from scattering

These effects are mainly low-frequency components.

Therefore, a **high-pass homomorphic filter** is preferred because it:

- Suppresses illumination variations.
- Reduces underwater haze.
- Enhances object boundaries.
- Increases local contrast.
- Emphasises coral textures and structural details.

Step 3: Selection of Cutoff Frequency

The cutoff frequency D_0controls the separation between illumination and reflectance components.

Very Low Cutoff Frequency

Effects

- Excessive removal of low frequencies.
- Loss of important object details.
- Over-enhanced image appearance.

Very High Cutoff Frequency

Effects

- Insufficient illumination correction.
- Remaining haze and poor contrast.

Optimal Cutoff Frequency

An intermediate cutoff frequency is selected to:

- Remove uneven illumination.
- Preserve coral structures.
- Retain significant image details.
- Avoid excessive sharpening.

Justification

Since underwater degradation primarily affects large-scale brightness variations, a moderate cutoff frequency effectively removes unwanted low-frequency illumination while preserving useful object information.

Step 4: Noise Suppression

Enhancement procedures may amplify noise. Therefore, filtering is performed before segmentation.

Median Filtering

Median filtering effectively removes impulse noise while preserving edges.

Advantages

- Eliminates isolated noisy pixels.
- Retains coral boundaries.
- Preserves fine underwater structures.

Gaussian Smoothing

A Gaussian filter can be applied to reduce high-frequency noise:

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Benefits

- Smooths noise.
- Produces cleaner segmentation results.
- Improves region uniformity.

Step 5: Object Segmentation

After enhancement, segmentation is performed to isolate coral regions and underwater objects.

Method 1: Adaptive Thresholding

When intensity differences exist between corals and the background, adaptive thresholding can be applied.

Thresholding Rule

$$g(x,y) = \begin{cases} 1, \& f(x,y) > T(x,y) \\ 0, \& f(x,y) \leq T(x,y) \end{cases}$$

where:

- $T(x,y)$is a local threshold.

Advantages

- Handles uneven illumination.
- Works effectively in varying underwater conditions.
- Extracts foreground objects accurately.

Contribution

Adaptive thresholding generates a binary image where corals and submerged structures appear as foreground regions.

Method 2: Region-Based Segmentation (Recommended)

For complex underwater scenes, region growing provides more reliable results.

Procedure

1. Select seed points within coral regions.
2. Examine neighbouring pixels.
3. Merge pixels with similar intensity or texture.
4. Continue until the stopping criterion is satisfied.

Advantages

- Produces connected object regions.
- Preserves coral shapes.
- Handles intensity variations.
- Provides better object isolation.

Contribution

Region growing extracts complete coral formations rather than fragmented pixel groups.

- **Step 6: Morphological Processing**

After segmentation, morphological operations refine the results.

Opening Operation

$$A \circ B$$

Purpose

- Removes small unwanted regions.
- Eliminates isolated noise pixels.

Closing Operation

$$A \cdot B$$

Purpose

- Fills small holes within coral regions.
- Connects nearby fragmented structures.

Contribution

Morphological processing improves object continuity and produces cleaner segmented regions.

Importance of the Order of Operations

The sequence of processing significantly affects segmentation accuracy.

Why This Order Is Important?

1. Enhancement Before Segmentation

- Raw underwater images contain haze and poor contrast.
- Segmenting directly would produce inaccurate boundaries.
- Enhancement increases object visibility.

2. Homomorphic Filtering Before Thresholding

- Removes illumination variations.
- Improves intensity consistency.
- Makes object-background separation easier.

3. Noise Reduction Before Segmentation

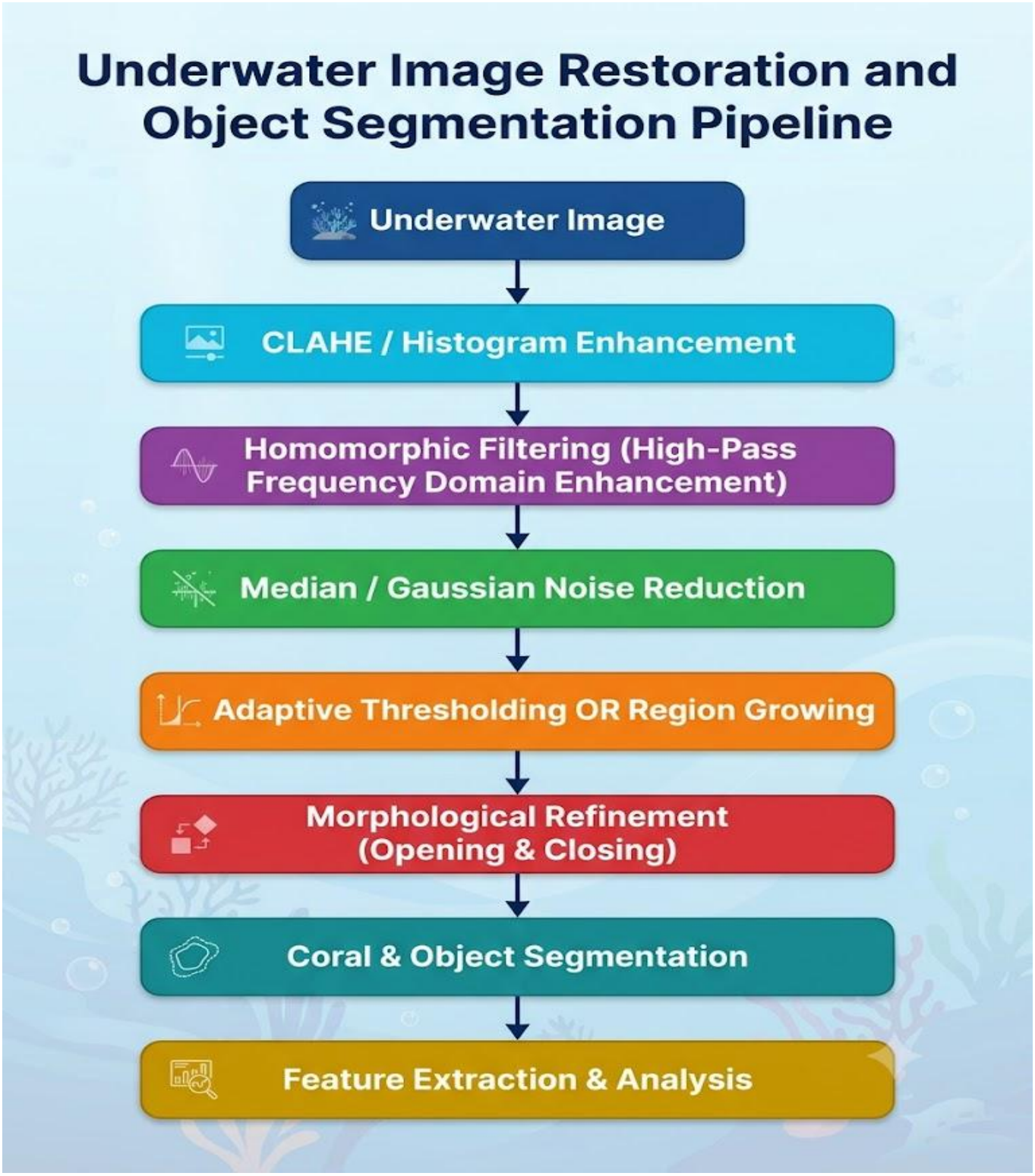
- Prevents false object detection.
- Improves region homogeneity.

4. Morphological Processing After Segmentation

- Refines detected regions.
- Removes small artefacts.
- Produces smooth object boundaries.

Changing this order may result in fragmented coral regions, false detections, and reduced segmentation accuracy.

Complete Processing Pipeline



4. Fingerprint Image Enhancement and Ridge Extraction

A biometric security system captures fingerprint images using optical sensors. The images exhibit smudges, uneven pressure, dry or wet regions, and noise, resulting in broken or merged ridge patterns. The system must extract clear ridge structures for subsequent matching.

Question: Apply suitable grey-level transformations, histogram equalisation, and spatial filtering to improve ridge clarity. Use thresholding and edge or line detection to extract ridge patterns. Apply edge linking and boundary detection to obtain continuous ridge contours. Present the processing sequence and explain the role of each operation in obtaining a usable ridge map.

Fingerprint Image Enhancement and Ridge Extraction

Introduction

Fingerprint recognition is one of the most widely used biometric authentication techniques because every individual possesses a unique ridge-valley pattern. However, fingerprint images acquired using optical sensors often suffer from various quality issues such as smudges, uneven finger pressure, dry or wet skin conditions, dirt, sensor noise, and poor contrast. These degradations can cause ridge discontinuities, merged ridges, broken ridge structures, and unclear minutiae points, which reduce matching accuracy.

To obtain a reliable fingerprint representation, image enhancement techniques such as grey-level transformations, histogram equalisation, and spatial filtering are applied before ridge extraction. Subsequently, thresholding, edge or line detection, edge linking, and boundary detection are used to obtain continuous ridge contours suitable for feature extraction and fingerprint matching.

Step 1: Grey-Level Transformation

The first stage aims to improve the visibility of ridge structures by enhancing image contrast.

Contrast Stretching

Fingerprint images often occupy only a small portion of the available intensity range. Contrast stretching expands the grey-level range using:

$$s = \frac{(r - r_min)}{(r_max - r_min)}(L - 1)$$

where:

- r= input pixel intensity
- s= output intensity
- L= number of grey levels

Purpose

- Increases distinction between ridges and valleys.
- Makes faint ridge patterns more visible.
- Improves ridge continuity.

Gamma Correction

The transformation is given by:

$$s = cr^{\gamma}$$

Effect

- $\gamma < 1$: brightens dark fingerprint regions.
- $\gamma > 1$: suppresses excessively bright areas.

Contribution

Gamma correction compensates for dry or wet fingerprint regions and produces more uniform ridge visibility.

- **Step 2: Histogram Equalisation**

After contrast stretching, histogram equalisation is applied to redistribute intensity values over the entire grey-scale range.

Histogram Equalisation

$$s = T(r)$$

where $T(r)$is the cumulative probability distribution function.

Benefits

- Improves overall image contrast.
- Enhances weak ridge structures.
- Increases separation between ridges and valleys.

- Reveals hidden details in low-quality fingerprints.

Adaptive Histogram Equalisation (CLAHE)

For fingerprints with non-uniform pressure distribution, CLAHE is preferred.

Advantages

- Performs local contrast enhancement.
- Preserves ridge details.
- Prevents over-amplification of noise.
- Enhances ridges in both bright and dark regions.

Contribution

Produces a more uniform fingerprint image suitable for subsequent filtering and segmentation.

- **Step 3: Spatial Filtering**

Even after enhancement, fingerprint images may contain noise and irregular artifacts.

Median Filtering

Median filtering replaces each pixel by the median value of neighbouring pixels.

$$f(x, y) = Median(N)$$

Advantages

- Removes salt-and-pepper noise.
- Preserves ridge edges.
- Maintains structural details.

Gaussian Filtering

Gaussian filtering smooths random noise using:

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Benefits

- Reduces sensor noise.
- Produces smoother ridge patterns.
- Prevents false edge generation.

Sharpening Filter

Laplacian sharpening may be applied:

$$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

Contribution

- Enhances ridge boundaries.
- Improves ridge-valley differentiation.
- Highlights fine fingerprint structures.

Step 4: Thresholding

The enhanced image must be converted into a binary representation.

Adaptive Thresholding

Since fingerprint images often contain non-uniform illumination and pressure variations, adaptive thresholding is preferred.

$$g(x,y) = \begin{cases} 1, \& f(x,y) > T(x,y) \\ 0, \& f(x,y) \leq T(x,y) \end{cases}$$

where:

- $T(x,y)$is a local threshold.

Benefits

- Separates ridges from valleys.
- Handles varying fingerprint intensity.
- Produces more accurate ridge maps.

Contribution

Creates a binary image in which ridges become foreground regions while valleys form the background.

- **Step 5: Ridge Extraction Using Edge or Line Detection**

Fingerprint ridges are essentially line structures. Therefore, line and edge detection techniques are highly effective.

Line Detection

Common masks:

Horizontal Ridge Detector

$$\begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

Vertical Ridge Detector

$$\begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$$

Contribution

- Enhances ridge orientation.
- Extracts ridge structures.
- Highlights ridge continuity.

Edge Detection

Canny Edge Detector

Processing stages:

1. Noise suppression
2. Gradient computation
3. Non-maximum suppression
4. Double thresholding
5. Edge tracking

Gradient magnitude:

$$M = \sqrt{G_x^2 + G_y^2}$$

Advantages

- Detects ridge boundaries accurately.
- Produces thin edges.
- Provides high localisation accuracy.

Contribution

Identifies ridge contours and ridge-valley transitions effectively.

- **Step 6: Edge Linking**

After edge detection, ridge contours may remain fragmented because of noise or poor image quality.

Hysteresis Linking

Used within the Canny algorithm.

Benefits

- Preserves connected ridge structures.
- Eliminates isolated false edges.

- Extends weak ridge segments connected to strong edges.

Morphological Dilation

$$A \oplus B$$

Benefits

- Connects broken ridge fragments.
- Restores interrupted ridge patterns.
- Improves ridge continuity.

Contribution

Produces continuous ridge networks suitable for fingerprint analysis.

- **Step 7: Boundary Detection**

Boundary detection extracts complete ridge contours and helps identify fingerprint features.

Contour Tracing

Boundary pixels are traced to form complete ridge outlines.

Information Obtained

- Ridge shape
- Ridge length
- Ridge orientation
- Ridge continuity

- **Ridge Skeletonisation (Optional)**

The ridge thickness is reduced to a single-pixel-wide representation.

Benefits

- Preserves ridge topology.
- Simplifies feature extraction.
- Facilitates minutiae detection.

Contribution

Provides a clean ridge map for fingerprint matching algorithms.

Role of Each Operation

Operation	Contribution
Contrast Stretching	Improves ridge visibility
Gamma Correction	Handles dry/wet regions
Histogram Equalisation	Enhances global contrast
CLAHE	Improves local ridge details
Median/Gaussian Filtering	Removes noise
Sharpening	Enhances ridge boundaries
Adaptive Thresholding	Separates ridges from valleys
Line Detection	Extracts ridge structures
Edge Detection	Locates ridge boundaries
Edge Linking	Connects fragmented ridges
Boundary Detection	Forms continuous ridge contours
Skeletonisation	Produces a usable ridge map

Why This Sequence is Important

1. Enhancement Before Extraction

- Poor-quality fingerprints contain weak ridge information.
- Contrast enhancement makes ridges clearly distinguishable.

2. Noise Removal Before Detection

- Noise produces false ridge segments.
- Filtering ensures accurate ridge extraction.

3. Thresholding Before Edge Analysis

- Converts the image into a clear ridge-valley pattern.
- Simplifies feature identification.

4. Edge Linking After Detection

- Connects fragmented ridge segments.
- Restores ridge continuity.

5. Boundary Detection at the End

- Works best when clean connected ridges already exist.
- Produces accurate ridge contours.

To obtain a usable fingerprint ridge map, the image is first enhanced using **grey-level transformations and histogram equalisation** to improve ridge visibility. **Median and Gaussian filtering** remove noise, while **Laplacian sharpening** enhances ridge boundaries. **Adaptive thresholding** separates ridges from valleys, after which **line detection and Canny edge detection** extract ridge structures. **Edge linking and morphological operations** connect fragmented ridges, and **boundary detection with skeletonisation** generates a continuous one-pixel-wide ridge map. This processing pipeline significantly improves fingerprint quality and enables reliable minutiae extraction and biometric matching.

5. Retinal Fundus Image Analysis for Blood Vessel and Lesion Extraction

An ophthalmology clinic uses fundus cameras to screen for diabetic retinopathy. The images show uneven illumination, a bright optic disc, low-contrast thin blood vessels, and small bright or dark lesions (such as exudates and haemorrhages) mixed with noise.

Question: Apply suitable contrast enhancement and an appropriate smoothing filter to improve the visibility of vessels and lesions while preserving fine structures. Use edge and line detection to identify vessel candidates, and thresholding or region-based segmentation to isolate lesion regions. Apply edge linking and boundary detection to obtain continuous vessel paths and lesion outlines. Explain the processing sequence and the contribution of each technique.

Retinal Fundus Image Analysis for Blood Vessel and Lesion Extraction

Introduction

Retinal fundus imaging is widely used for the early detection and diagnosis of eye diseases such as diabetic retinopathy, glaucoma, and hypertensive retinopathy. A typical fundus image contains important anatomical structures such as the optic disc, macula, blood vessels, and retinal background. In diabetic retinopathy screening, clinicians are especially interested in detecting thin blood vessels and lesions such as exudates, microaneurysms, and haemorrhages.

However, fundus images often present challenges including low-contrast blood vessels, bright optic disc regions, small bright or dark lesions, uneven illumination, and image noise. Therefore, a sequence of image enhancement, filtering, vessel detection, lesion segmentation, edge linking, and boundary extraction techniques is required to accurately identify retinal abnormalities while preserving delicate vessel structures.

Step 1: Contrast Enhancement

The first step is to improve the visibility of retinal structures, particularly thin blood vessels and small lesions.

Contrast Stretching

Contrast stretching expands the intensity range of the image:

$$s = \frac{(r - r_min)}{(r_max - r_min)}(L - 1)$$

where:

- r= input intensity
- s= output intensity
- L= number of grey levels

Purpose

- Enhances weak vessel structures.
- Improves visibility of small lesions.
- Increases vessel-to-background contrast.
- Makes pathological regions easier to distinguish.

Adaptive Histogram Equalisation (CLAHE)

CLAHE is preferred because retinal images often have local intensity variations.

Advantages

- Enhances local vessel contrast.
- Improves visibility of microvascular structures.

- Preserves fine retinal details.
- Prevents excessive amplification of noise.

Contribution

Thin vessels that are barely visible in the original image become more distinguishable after CLAHE enhancement.

Step 2: Image Smoothing and Noise Reduction

After contrast enhancement, noise must be reduced without destroying important retinal features.

Median Filtering

Median filtering replaces a pixel value with the median of neighbouring pixels.

$$f(x, y) = Median(N)$$

Advantages

- Removes impulse noise.
- Preserves vessel edges.
- Maintains lesion boundaries.

Contribution

Reduces small imaging artefacts while retaining fine vascular details.

Gaussian Filtering

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Advantages

- Reduces random sensor noise.
- Smooths background variations.
- Enhances segmentation stability.

Contribution

Produces a cleaner retinal image for vessel and lesion detection.

- **Step 3: Blood Vessel Detection Using Edge and Line Detection**

Blood vessels appear as elongated line-like structures. Therefore, line and edge detection methods are highly effective.

Line Detection

Directional line masks highlight vessel segments in different orientations.

Horizontal Line Mask

$$\begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

Vertical Line Mask

$$\begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$$

Contribution

- Enhances vessel candidates.
- Detects thin vascular structures.
- Identifies vessel continuity.

- Improves visibility of weak capillaries.

Edge Detection

Canny Edge Detector

The Canny operator is widely used because it can accurately detect weak vessel boundaries.

Processing Stages

1. Noise suppression.
2. Gradient calculation.
3. Non-maximum suppression.
4. Double thresholding.
5. Edge tracking.

Gradient magnitude:

$$M = \sqrt{G_x^2 + G_y^2}$$

Advantages

- Detects thin vessel boundaries.
- Produces precise edges.
- Reduces false detections.
- Maintains vessel continuity.

Contribution

Clearly identifies blood vessel outlines and lesion boundaries.

Step 4: Lesion Segmentation

Lesions such as exudates and haemorrhages must be separated from healthy retinal tissue.

Thresholding

Adaptive Thresholding

$$g(x,y) = \begin{cases} 1, \& f(x,y) > T(x,y) \\ 0, \& f(x,y) \leq T(x,y) \end{cases}$$

where:

- $T(x,y)$is a local threshold.

Advantages

- Handles illumination changes.
- Extracts bright exudates effectively.
- Detects dark haemorrhage regions.

Contribution

Generates an initial lesion map for further analysis.

Region-Based Segmentation

For more accurate lesion extraction, region growing can be used.

Procedure

1. Select seed points within lesion regions.
2. Examine neighbouring pixels.
3. Merge pixels with similar intensity values.
4. Continue until no similar neighbours remain.

Advantages

- Extracts complete lesion regions.
- Preserves lesion shape.
- Reduces fragmented segmentation.

Contribution

Produces more accurate boundaries around pathological regions.

- **Step 5: Edge Linking**

Detected blood vessel segments may appear fragmented due to low contrast or noise.

Hysteresis Linking

Used within the Canny detector.

Benefits

- Preserves genuine vessel edges.
- Connects weak vessel segments.
- Eliminates isolated false edges.

Morphological Dilation

$$A \oplus B$$

Benefits

- Bridges small vessel gaps.
- Restores interrupted vascular paths.
- Connects neighbouring lesion boundaries.

Contribution

Ensures continuous vessel network representation.

- **Step 6: Boundary Detection**

Boundary extraction is performed to obtain complete vessel paths and lesion contours.

Contour Detection

Contour tracing identifies the outer boundary of segmented objects.

Information Obtained

- Vessel length
- Vessel width
- Lesion perimeter
- Lesion area
- Shape characteristics

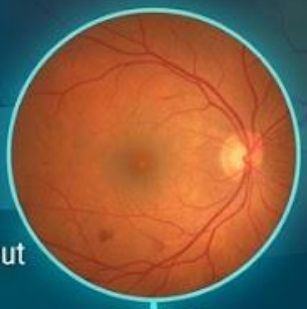
Contribution

Provides quantitative measurements for medical analysis.

Retinal Fundus Image Analysis for Blood Vessel and Lesion Extraction

Input Retinal Fundus

Input Retinal Fundus Image



Color Fundus Input

Green Channel

Green Channel & Enhancement
Green Channel Extraction
& Contrast Enhancement

Green Channel



Vessel Segmentation

Matched Filtering & Vessel Segmentation

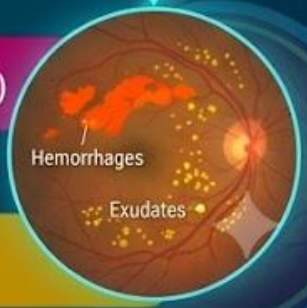
Vessel Map



Lesion Identification

Lesion Detection (Exudates & Hemorrhages)

Lesion Overlay



Feature Extraction

Connected Component Analysis
Connected regions are labelled and analysed.
Enables Detection Of

- Exudates
- Haemorrhages
- Microaneurysms
- Vessel branches
- Abnormal vascular structures

Contribution
Allows classification and measurement of disease-related features.

- **Contribution of Each Technique**

Technique	Contribution
Contrast Stretching	Improves vessel visibility
CLAHE	Enhances local vessel contrast
Median Filtering	Removes impulse noise
Gaussian Filtering	Suppresses random noise
Line Detection	Extracts vessel-like structures
Edge Detection	Detects vessel and lesion boundaries
Adaptive Thresholding	Separates lesions from background
Region Growing	Extracts complete lesion regions
Edge Linking	Connects fragmented vessels
Boundary Detection	Extracts vessel paths and lesion contours
Component Analysis	Measures retinal abnormalities

Why This Sequence is Important
1. Enhancement Before Detection

- Low-contrast vessels become clearly visible.
- Small lesions become easier to identify.

2. Noise Removal Before Segmentation

- Prevents false vessel detection.
- Improves lesion extraction accuracy.

3. Vessel Detection Before Edge Linking

- Initial vessel segments must first be identified.
- Edge linking then forms continuous vascular networks.

4. Segmentation Before Boundary Detection

- Boundaries can only be extracted after objects are separated from the background.

5. Boundary Detection at the Final Stage

- Provides accurate measurements of vessels and lesions.

- Supports clinical decision-making and disease grading.

For retinal fundus image analysis, **contrast stretching and CLAHE** are first applied to enhance the visibility of thin blood vessels and retinal lesions. **Median and Gaussian filtering** reduce noise while preserving delicate structures. **Line detection and Canny edge detection** are then used to identify vessel candidates and vessel boundaries. **Adaptive thresholding or region-based segmentation** isolates bright exudates and dark haemorrhages from the retinal background. **Edge linking and morphological operations** connect fragmented vessel segments and lesion contours, while **boundary detection and connected component analysis** provide accurate measurements of vessel networks and lesion regions. This complete processing pipeline enables reliable extraction of retinal features and supports the early diagnosis of diabetic retinopathy and other retinal diseases.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand